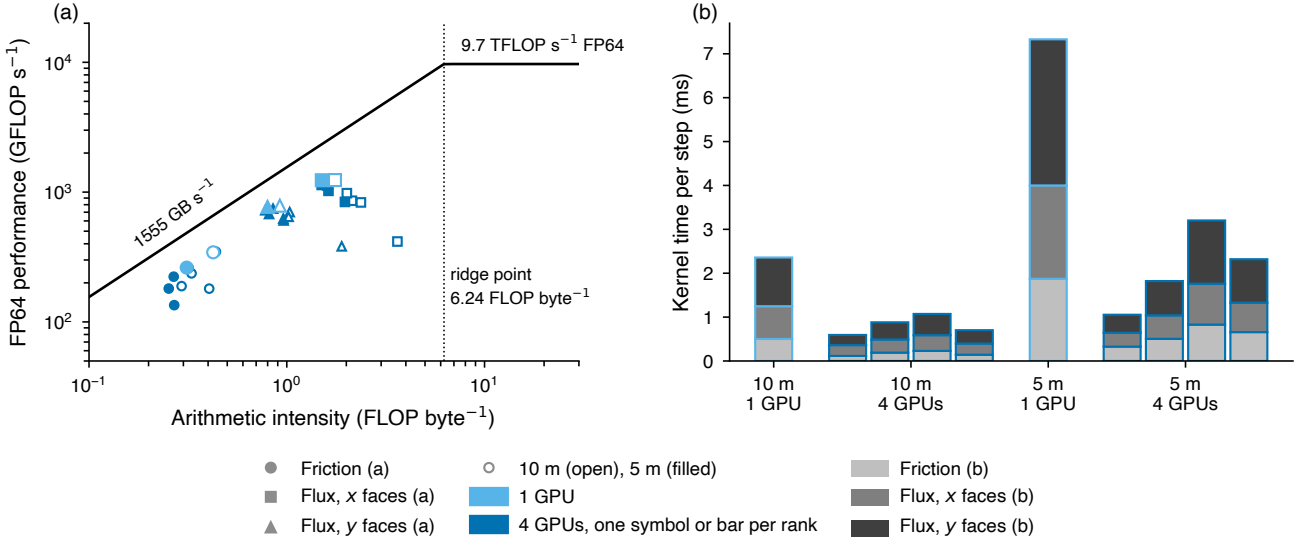
(a)
FP64 performance (GFLOP s^{-1})
10^4
10^3
10^2
9.7 TFLOP s^{-1} FP64
1555 GB s^{-1}
ridge point
6.24 FLOP byte^{-1}
10^{-1}
10^0
10^1
Arithmetic intensity (FLOP byte^{-1})
(b)
Kernel time per step (ms)
0
1
2
3
4
5
6
7
10 m
1 GPU
10 m
4 GPUs
5 m
1 GPU
5 m
4 GPUs
Friction (a)
Flux, x faces (a)
Flux, y faces (a)
10 m (open), 5 m (filled)
1 GPU
4 GPUs, one symbol or bar per rank
Friction (b)
Flux, x faces (b)
Flux, y faces (b)